Bits and Bytes of Networking

1. Networking Connectivity & DNS Diagnostic Tool

- **Candidate:** Mosala Stanley Kgadimonyane
- **Technical Stack:** Python 3, TCP/IP , ICMP (Ping), DNS
- **DATE:** March 2026

➢ Moving beyond local hardware, this project focuses on the. “Communication Layer”. I developed a Python utility to verify external connectivity, resolve domain names, and identify local network configuration.

- **Technical Verification Strategy.**
  To troubleshoot connectivity effectively, I designed this utility to test the Network Layer (Layer 3) and Application Layer (Layer 7) of the OSI Model. By isolating these components, a technician can quickly determine if a connectivity issue is physical configuration based, or a service provider failure.

- **Local loopback & IP Verification:** Confirms the local network stack is initialized and the device has a valid identity on the network.

- **ICMP Reachability (Ping):** Tests the Gateway by sending packets to a known external server (8.8.8.8). This confirms that data can physically leave the local network and receive a response.

- **DNS Resolution Analysis:** Verifies that the Phonebook of the internet is working by translating the domain google.com into a machine readable IP address.

- **<u>Key Metrics Monitored</u>**

Test	Objective	Success Criteria
Ping Statistics	Measure latency & stability	0% Packet Loss
DNS Lookup	Verify Name Resolution	Valid IP returned for Hostname
Interface Check	Confirm Local Connectivity	Active IP Address detected

```
localhost:~/Portfolio/The_Bits_and_Bytes_of_Networking#
localhost:~/Portfolio/The_Bits_and_Bytes_of_Networking#l
s
net_test.py          networking_notes.py
localhost:~/Portfolio/The_Bits_and_Bytes_of_Networking#
python3 net_test.py
--- NETWORK CONNECTIVITY DASHBOARD ---
Device Name: localhost
Local IP Address: 127.0.0.1

Testing connection to Google DNS (8.8.8.8)...
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: seq=0 ttl=118 time=30.851 ms
64 bytes from 8.8.8.8: seq=1 ttl=118 time=20.523 ms
64 bytes from 8.8.8.8: seq=2 ttl=118 time=39.365 ms

--- 8.8.8.8 ping statistics ---
3 packets transmitted, 3 packets received, 0% packet los
s
round-trip min/avg/max = 20.523/30.246/39.365 ms
Result: Success! Network is reachable.

Testing DNS Resolution (google.com)...
localhost:~/Portfolio/The_Bits_and_Bytes_of_Networking#
```

- **<u>Results and Analysis: Connectivity Verified</u>**
  Based on the terminal output generated by the Python diagnostic utility, the network environment is confirmed to be stable and fully functional across the relevant OSI layers.

- The ICMP test turned 0% packet loss with an average round trip time (RTT) of 30.246ms. This confirms that the path between the local device and the external gateway is clear of congestion or hardware interference.

- The DNS resolution test successfully mapped google.com to its current IP address. This confirms that the Application Layer (Layer 7) can communicate with DNS servers to translate human readable names into machine readable addresses

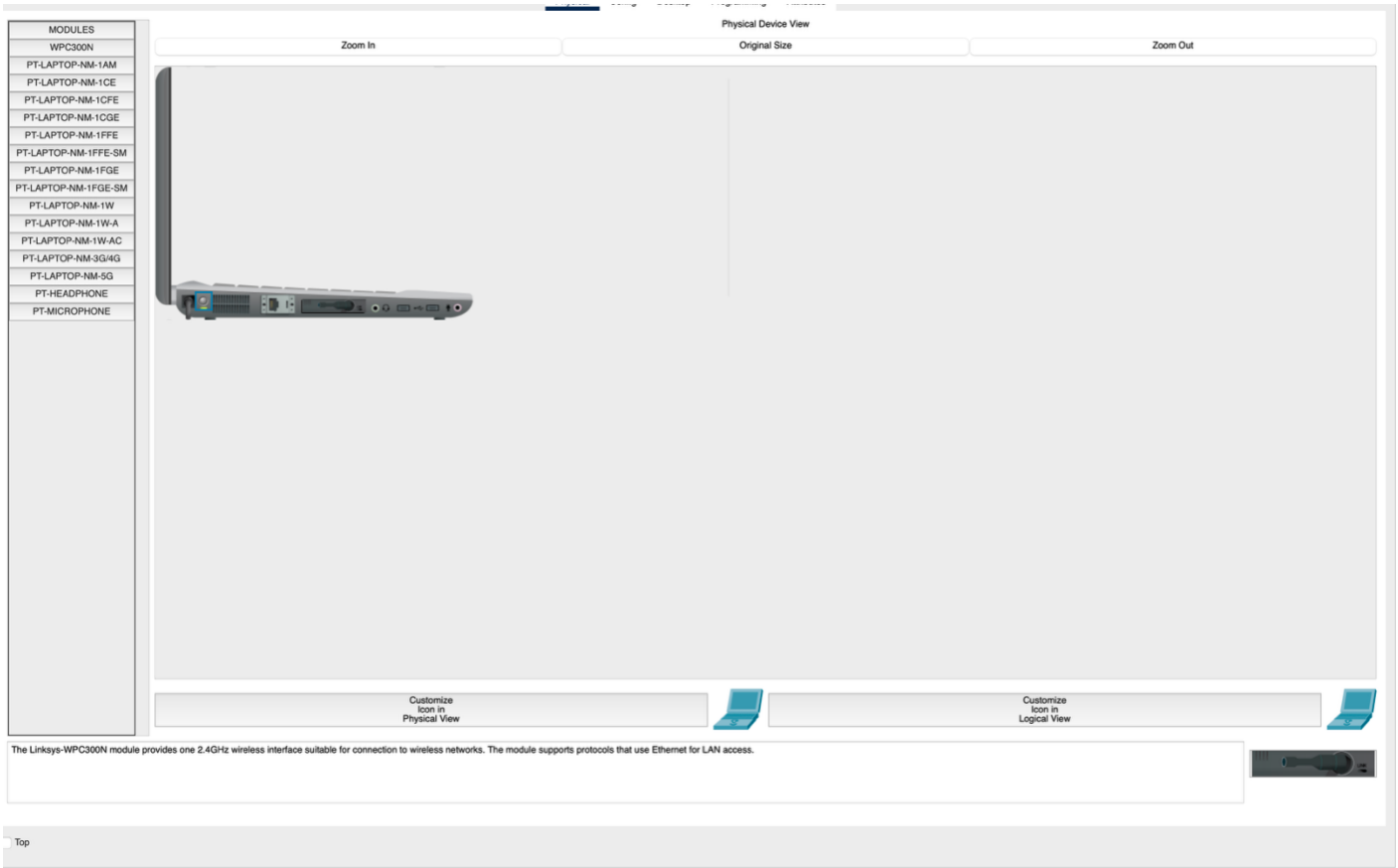
- The device has a valid local identity and a functional bridge to the wider internet. No further escalation is required at this time.

2. Hardware Adaptation and Dynamic Addressing

- **<u>Hardware Adaptation</u>**

To support mobile connectivity within the topology, the physical layer of the Laptop node was modified.

- Performed a Hot Swap of the Network Interface Card (NIC).
- Replaced the standard copper RJ-45 module with the WPC300N Wireless Module.
- Transitioning from 802.3 (Wired) to 802.11 (Wireless) protocols to simulate modern flexible workspace.
- Before configuring the logical network setting, the Laptop required a physical hardware modification transition from a wired (802.3) connection to a wireless (802.11) interface.

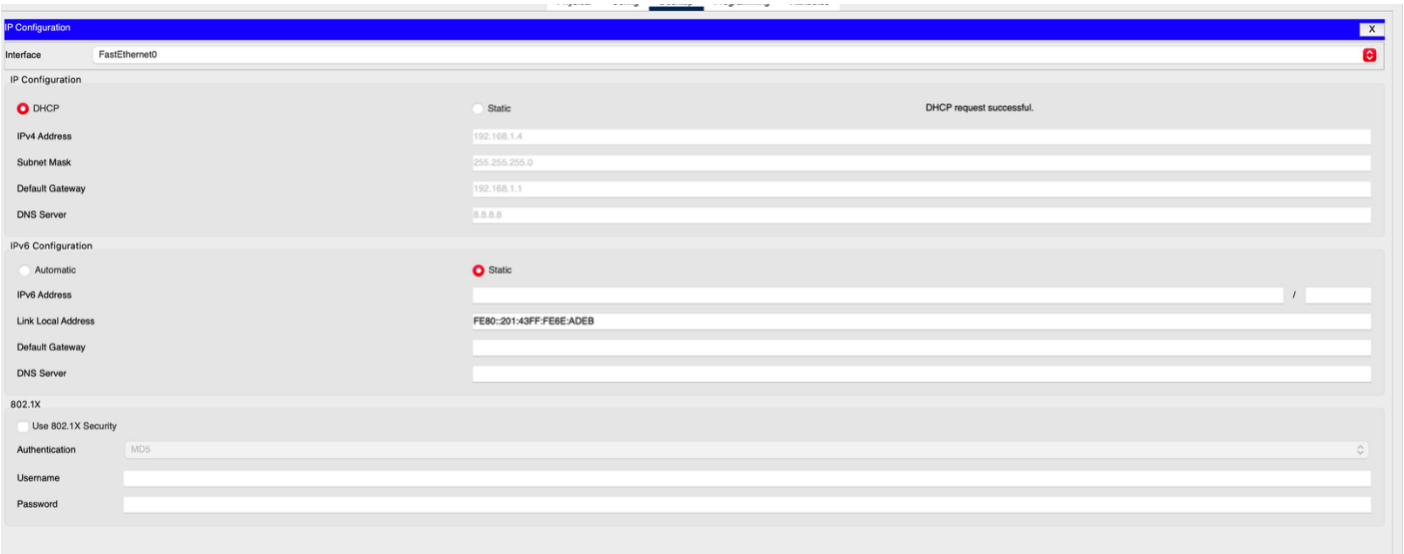


- **<u>Logical Addressing & DHCP Implementation</u>**

Following the physical hardware modification, the logical layer was configured to ensure the device could communicate with the rest of the network. To move beyond manual static entry, the network was transitioned to a dynamic model.

- The core ISR4331 Router was configured to act as a centralized DHCP server.This ensures that any device connecting to the HomeNetwork SSID wired or wireless automatically receives a valid IP address, subnet mask and gateway.

- As shown in the following configuration data, the Laptop successfully requested and received a unique identity within the 192.168.1.0/24 subnet.



- **<u>Technical Analysis of Results</u>**

- **Layer 2 Connectivity:** The Laptop has successfully associated with the Wireless Access Point.

- **Layer 3 Services:** The router successfully processed the DHCPDISCOVER broadcast and assigned the IP 192.168.1.4.

- **Default Gateway Awareness:** The device is now aware to the exit point (192.168.1.1), enabling traffic to flow outside the local segment.

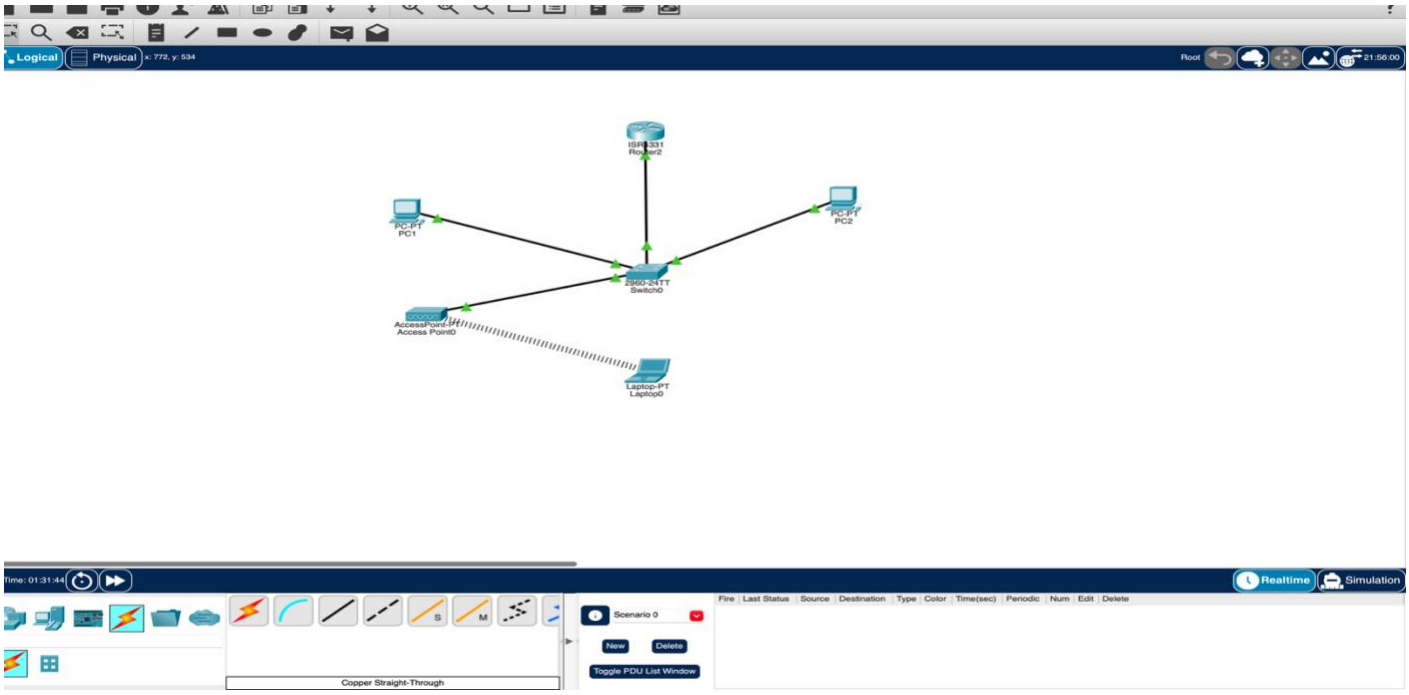
3. Final Network Verification and Topology Overview

- With the hardware adapted and the logical addressing successfully implemented via DHCP, the full hybrid network is now operational. This topology demonstrates a scalable Branch Office model, integrating standard workstations with mobile wireless assets.

- **Wired Segment:** Stable connectivity between PC1, PC2, and centralized switch.

- **Wireless Segment:** Successful Layer 2 association between the Laptop and the Access Point.

- **Core Connectivity:** All segments successfully route through ISR4331 Gateway for external service simulation.



- As illustrated in the topology map, all link lights are green, signifying active physical and data link layer connections. The dashed line between the Access Point and the laptop confirms a successful 802.11 wireless handshake. This completed environment provides the necessary infrastructure for end-to-end communication testing including DNS Resolution and web traffic simulation.

- Through this lab, I have successfully demonstrated the ability to design, configure, and troubleshoot a hybrid network environment. I managed hardware transitions, implemented automated IP addressing (DHCP), and verified connectivity using industry standard diagnostic tools. This project confirms my readiness to move into advanced network security and server hardening modules